

# Take home Assessment

## 1) Source code

<https://github.com/Libish77/todo-assesment>

## 2) Read me

### a) How to run the solution

Prerequisites: .NET 10 SDK, Node.js (LTS), Visual Studio 2026, VS Code.

#### 1. Backend (Visual Studio)

- Open CleanArchitecture.sln (or the solution in the repo root).
- Set Web (src/Web) as the startup project.
- Press F5 to run the API.
- The API runs at `https://localhost:5001` and/or `http://localhost:5000` (see `launchSettings.json`).
- On first run, the app creates the SQLite database (`CleanArchitecture.db` under `src/Web`) via `EnsureCreatedAsync()`.

#### 2. Frontend (VS Code)

- Open `src/Web/ClientApp` in VS Code.
- Run `npm install` once if you have not already.
- Run `ng serve` (or `npm start`).
- Open the Angular app (typically `http://localhost:4200`).
- `proxy.conf.js` forwards `/api` and related paths to the API at `https://localhost:5001`.

#### 3. Sign in

- Use the seeded account if prompted: `administrator@localhost` / `Administrator1!`
- Or you can register with your email and password as well.
- Go to Tasks in the nav menu to use the todo feature.

#### Tests:

- `dotnet test tests/Application.FunctionalTests/Application.FunctionalTests.csproj`

### b) What was changed?

Domain: Added optional `DueDate` (`DateTime?`) on `TodolItem`.

Application: Extended `CreateTodolItemCommand` and `UpdateTodolItemDetailCommand` to accept `DueDate`; mapped `DueDate` in `TodolItemDto` (`GetTodos` query).

Validation: `CreateTodolItemCommandValidator` — title required; due date optional; due date cannot be in the past when provided.

Infrastructure: EF persists `DueDate` via existing `EnsureCreated` setup (no separate migration files).

Angular UI (`todo.component`):

Optional due date when creating an item (inline date input).

Due date in the item details dialog (edit).  
Due date shown per item in the list (— when empty).  
Status filter: All, Active, Completed (client-side).  
Overdue incomplete items: red styling and Overdue · prefix before the date.  
Tests: Added ShouldCreateTodoItemWithDueDate in CreateTodoItemTests.

c) Assumptions made

Filtering (All / Active / Completed) is done in the Angular client on the loaded list.

Overdue means: has a due date, due date is before today, and the item is not completed.

New items cannot be created with a past due date (API validation + HTML min on date inputs).

Existing items can become overdue as calendar days pass.

The template's SQLite + EnsureCreated approach is kept instead of adding EF Core migrations.

Default login uses the template's seeded administrator account.

d) Known limitations

No EF Core migration files; schema changes rely on EnsureCreated or deleting the local SQLite file.

UpdateTodoItemDetail does not have a FluentValidation validator for DueDate (create path is validated; edit relies mainly on UI min date).

No server-side filter API — filters only apply in the browser for the current list.

e) What I would improve with more time

Add UpdateTodoItemDetailCommandValidator (and tests) for due date rules on edit.

Add an Overdue (and/or “Due today”) filter in the UI.

Add EF Core migrations (or documented migration scripts) instead of only EnsureCreated.

Seed sample data with varied due dates (including one overdue) for easier demos.

Sort by due date and optional filter by priority.

### 3) Demo Instruction

1. Creating a to-do item with a due date

Select a todo list in the left sidebar (e.g. Tasks).

Click the row to add a new item (checkbox row at the bottom).

Enter a title.

Optionally choose a due date in the date field (today or a future date).

Press Enter or confirm to save.

The new item appears in the list with the due date on the right.

2. Editing a due date

Click the ... button on an existing item.

In the details dialog, set or change Due date, or clear the field to remove it.

Save/close the dialog.

The list updates with the new due date (or — if cleared).

3. Viewing due dates in the UI

Each item shows a due date on the right of the row (formatted, e.g. May 21, 2026).

Items without a due date show —.

4. Filtering active and completed items

Above the item list, open the filter dropdown.

Choose:

All — every item in the list

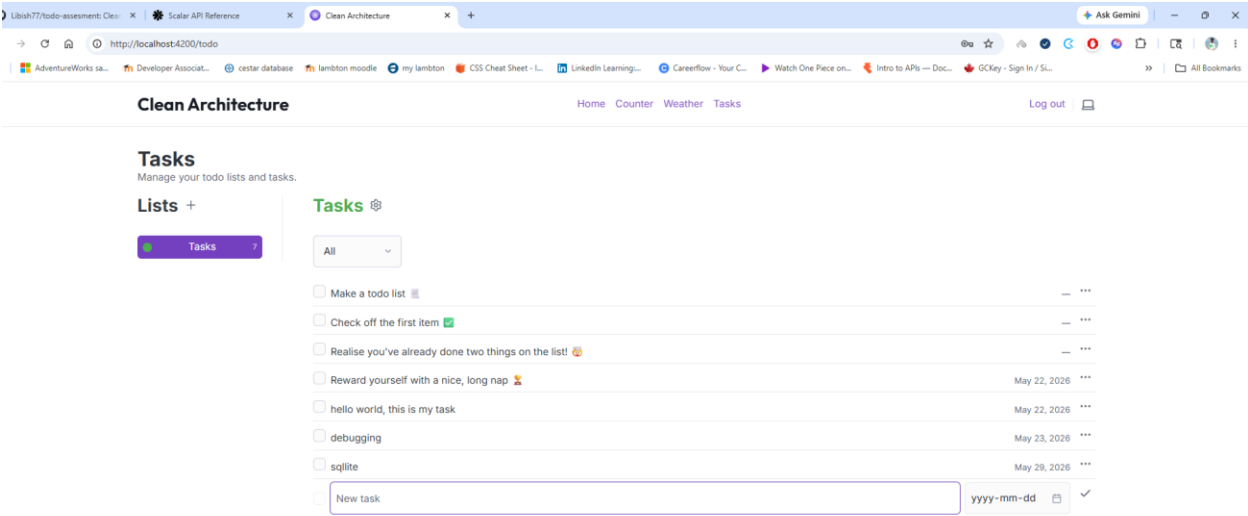
Active — items not checked / not done

Completed — checked / done items only

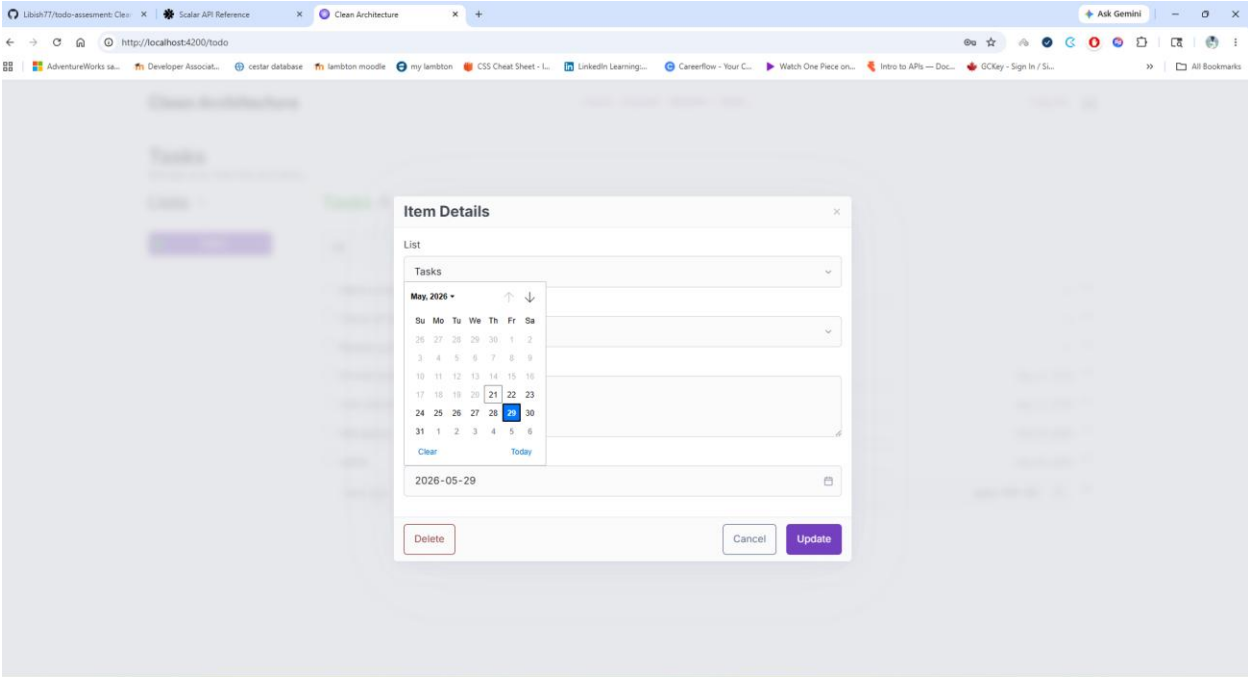
Confirm the list updates immediately (filtering is client-side).

5. Identifying overdue items

An item is overdue when it has a due date before today, is not completed, and is still open.



Add



Edit

Libish77/todo-assesment: Clea...

Scalar API Reference

Clean Architecture

+

Ask Gemini

http://localhost:4200/todo

AdventureWorks sa...Developer Associat...cestar databaseiamblon moodlemy lamblonCSS Cheat Sheet - L...LinkedIn Learning...Careerflow - Your C...Watch One Piece on...Intro to APIs --- Doc...GCKey - Sign In / Si...All Bookmarks

Clean Architecture

HomeCounterWeatherTasks

Log out

Tasks

Manage your todo lists and tasks.

Lists +

Tasks

Tasks

All

☐ Make a todo list

...

...

☐ Check off the first item

✓

...

☐ Realise you've already done two things on the list!

😂

...

☐ sd

Overdue - May 19, 2026

...

☐ hello world, this is my task

May 22, 2026

...

☐ debugging

May 23, 2026

...

☐ sqllite

May 29, 2026

...

☐ New task...

List